

# NGS Result Report

|                                               |                               |
|-----------------------------------------------|-------------------------------|
| Report ID: stage101-no_phenotype_match-report | Input allele: 5               |
| Genome build: GRCh38                          | Report state: Confirmed       |
| Generated: 2026-08-11                         | Template: professor-report-v1 |

## Clinical Features:

Accepted features: Neurologic abnormality (HP:0000707); Seizure (HP:0001250). Phenotype concordance: no supported association. No supported association was retained between the accepted phenotype and this synthetic allele.

## Method:

Allele-level evidence synthesis for an already filtered variant; this application did not perform sequencing or genome-wide prioritization. Call quality — QUAL: 99; FILTER: PASS; state: passed.

## Variant and source classification context:

**GENE1:c.101C>T p.(Arg34Trp)**

**Heterozygous**

***System classification: Not independently determined***

*Source: No independent application classification*

## Brief Interpretation(s):

The allele remains interpretable, but the retained evidence does not support an association with the accepted phenotype. Variant-level population, classification, and computational context remains source attributed and must not be converted into a phenotype match.

*Decision-support output for educational and research use; qualified human review is required.*

## Main Finding(s) in Detail:

| Gene & transcript         | Variant                            | Population frequency         |                            |                                                                 | Associated disease & ID                                                                                                                                                        | Zygosity              |
|---------------------------|------------------------------------|------------------------------|----------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|                           |                                    | gnomAD<br>gnomAD<br>r4       | 1000<br>Genomes<br>Phase 3 | Ensembl<br>Variation<br>on<br>GRCh38<br>variation<br>(fallback) |                                                                                                                                                                                |                       |
| GENE1<br>NM_000001.1      | c.101C>T<br>p.(Arg34Trp)           | 1.230e-05                    | No exact<br>record         | 2.000e-05                                                       | Synthetic<br>neurodevelopmental<br>condition<br>(MONDO:0000001)                                                                                                                | Heterozygous          |
| SIFT                      | PolyPhen-2                         | CADD                         |                            | Stable variant<br>ID                                            | Classification context                                                                                                                                                         | Inheritance           |
| deleterious<br>score 0.01 | probably<br>damaging<br>score 0.97 | elevated score<br>score 24.1 |                            | fixture ID:<br>SYNTHETIC-<br>VARIANT-<br>001                    | NCBI ClinVar: Uncertain<br>significance (VUS)<br>(criteria provided, single<br>submitter)<br><br>GeneBe: Automated<br>uncertain significance<br>(automated annotation<br>only) | autosomal<br>dominant |

Population values are exact-allele observations from the named source. Computational predictions are source-attributed and are not independent ACMG classifications. Missingness is written explicitly.

## Variant interpretation:

### GENE1:c.101C>T p.(Arg34Trp)

The allele remains interpretable, but the retained evidence does not support an association with the accepted phenotype. Variant-level population, classification, and computational context remains source attributed and must not be converted into a phenotype match.

### Variant(s) classification:

The retained sources provide uncertain-significance context without an independently adjudicated application classification. Reviewer-confirmed classification: Not available. ClinVar classification: Uncertain significance (VUS). Automated classification context: GeneBe automated uncertain significance. Conflict status: none. Sources: NCBI ClinVar, GeneBe automated annotation. The application has not independently adjudicated an ACMG/AMP classification.

### Method:

The application validated the supplied allele identity, collected bounded source-attributed evidence, preserved provider availability and fallback provenance, and rendered the reviewer-approved interpretation without a new model request.

### Comments / scope:

All allele and phenotype values in this golden report are synthetic and reserved for layout validation. One population observation was supplied by the explicitly identified fallback source. The CSpec record is context only; no specification criteria were executed. This allele-level report supports qualified human review and does not provide a diagnosis, treatment recommendation, or testing directive.

### References:

1. Standards and guidelines for the interpretation of sequence variants. [PMID: 25741868](#)
2. ACMG and AMP sequence variant interpretation guidance. [DOI: 10.1038/gim.2015.30](#)
3. Sequence variant interpretation standards. [PMCID: PMC4544753](#)

### Data Sources:

| Source            | Capability                       | Status               | Record or dataset         | Method                   | Link                        |
|-------------------|----------------------------------|----------------------|---------------------------|--------------------------|-----------------------------|
| Ensembl VEP       | variant consequence              | Available            | SYNTHETIC-VARIANT-001     | exact allele query       | <a href="#">Open record</a> |
| NCBI ClinVar      | clinical classification          | Available            | synthetic-clinvar-context | exact allele query       | <a href="#">Open record</a> |
| GeneBe            | automated classification context | Available            | 2-100001-C-T              | exact allele query       | <a href="#">Open record</a> |
| ClinGen / GenCC   | gene disease context             | Available            | MONDO:0000001             | gene context query       | <a href="#">Open record</a> |
| ClinGen CSpec     | specification context            | Available            | synthetic-context-only    | released metadata lookup | <a href="#">Open record</a> |
| gnomAD            | population frequency             | Available            | gnomAD r4                 | exact allele query       | Not available               |
| Ensembl Variation | population frequency             | Available (fallback) | SYNTHETIC-VARIANT-001     | exact overlap query      | <a href="#">Open record</a> |
| Europe PMC        | literature retrieval             | Available            | MED:25741868              | identifier lookup        | <a href="#">Open record</a> |